

Gradle Project

Step 1:

Create a new Gradle project

To create a new Gradle project using the command line:

\$gradle init --type java-application

```
vaishnavi@vaishnavi-virtualbox:~/devops21$ mkdir MyGradleApp
vaishnavi@vaishnavi-virtualbox:~/devops21$ cd MyGradleApp/
vaishnavi@vaishnavi-virtualbox:~/devops21/MyGradleApp$ gradle init --type java-application
Starting a Gradle Daemon (subsequent builds will be faster)

BUILD SUCCESSFUL in 5s
2 actionable tasks: 2 executed
vaishnavi@vaishnavi-virtualbox:~/devops21/MyGradleApp$
```

This command creates a new Java application project with a sample **build.gradle** file.

Step 2:

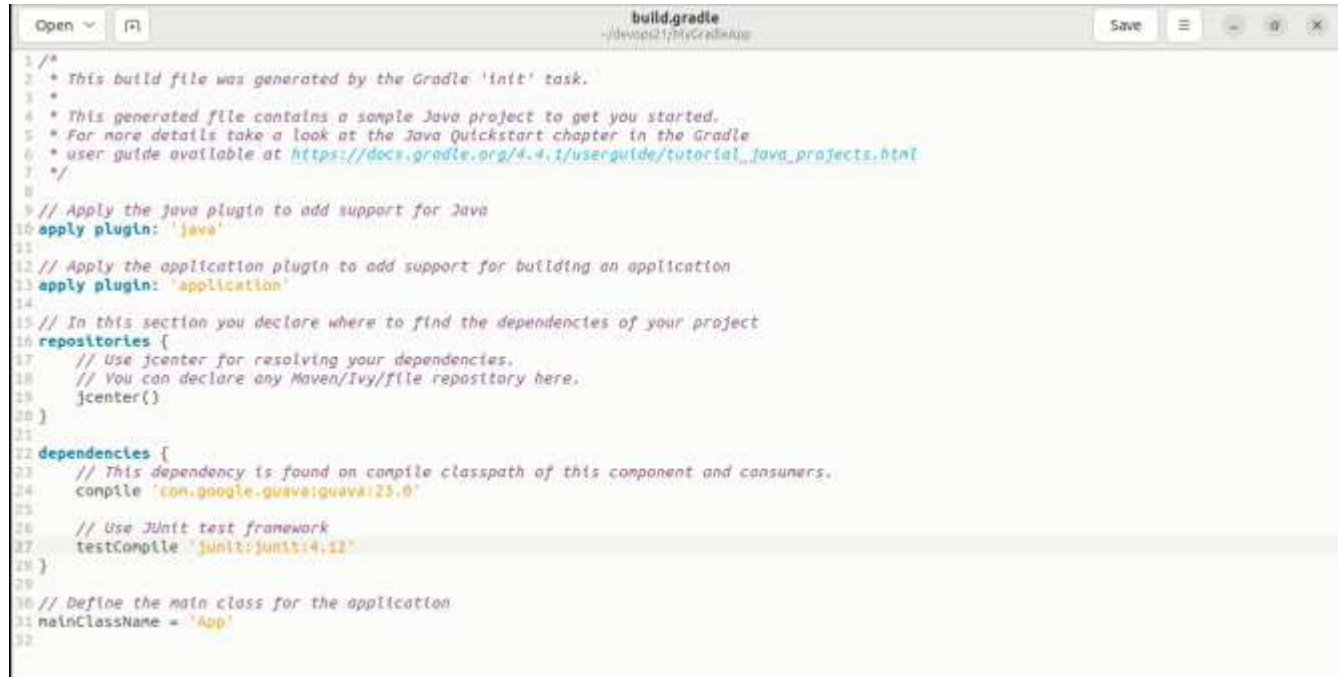
Execute \$tree command to view the project structure

```
vaishnavi@vaishnavi-virtualbox:~/devops21/MyGradleApp$ tree
.
├── build.gradle
├── gradle
│   └── wrapper
│       ├── gradle-wrapper.jar
│       └── gradle-wrapper.properties
├── gradlew
├── gradlew.bat
├── settings.gradle
└── src
    ├── main
    │   └── java
    │       └── App.java
    └── test
        └── java
            └── AppTest.java

7 directories, 8 files
vaishnavi@vaishnavi-virtualbox:~/devops21/MyGradleApp$
```

Step 3:

Edit the build.gradle file to add necessary plugins



```
1 /*
2  * This build file was generated by the Gradle 'init' task.
3  *
4  * This generated file contains a sample Java project to get you started.
5  * For more details take a look at the Java Quickstart chapter in the Gradle
6  * user guide available at https://docs.gradle.org/4.4.1/userguide/tutorial\_java\_projects.html
7  */
8
9 // Apply the java plugin to add support for Java
10 apply plugin: 'java'
11
12 // Apply the application plugin to add support for building an application
13 apply plugin: 'application'
14
15 // In this section you declare where to find the dependencies of your project
16 repositories {
17     // Use jcenter for resolving your dependencies.
18     // You can declare any Maven/Ivy/file repository here.
19     jcenter()
20 }
21
22 dependencies {
23     // This dependency is found on compile classpath of this component and consumers.
24     compile 'com.google.guava:guava:23.0'
25
26     // Use JUnit test framework
27     testCompile 'junit:junit:4.12'
28 }
29
30 // Define the main class for the application
31 mainClassName = 'App'
32
```

- **build.gradle** is a configuration file used by Gradle to build and manage a project
- **Plugins section**
 - java → enables Java compilation
 - application → allows running the project as an application
- **Repositories section**
 - Defines where dependencies are downloaded from
 - jcenter() → a repository for libraries
- **Dependencies section**
 - guava → provides utility functions for Java
 - junit → used for testing the application
- **Main Class**
 - mainClassName = 'App' → specifies the entry point of the program
- **Overall purpose**
 - Manages dependencies
 - Builds the project
 - Runs the application

Step 4:

Execute the given command to run the gradle project

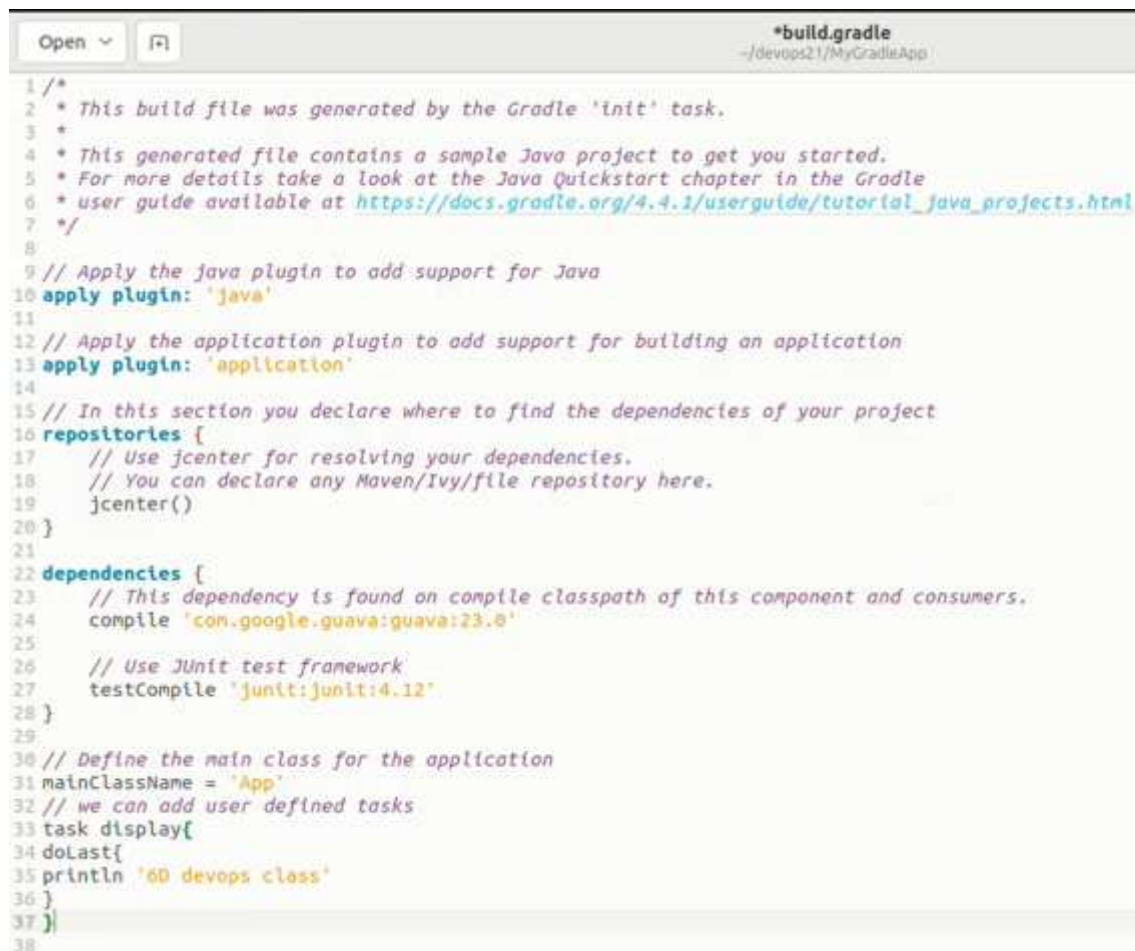
\$gradle run

```
vaishnavi@vaishnavi-virtualbox:~/devops21/MyGradleApp$ gradle run
> Task :run
Hello world.

BUILD SUCCESSFUL in 1s
2 actionable tasks: 1 executed, 1 up-to-date
vaishnavi@vaishnavi-virtualbox:~/devops21/MyGradleApp$
```

Step 5:

Add the user defined tasks to build.gradle at the end



The screenshot shows an IDE window titled '*build.gradle' with the file path '~/devops21/MyGradleApp'. The file contains the following Gradle build script:

```
1 /*
2  * This build file was generated by the Gradle 'init' task.
3  *
4  * This generated file contains a sample Java project to get you started.
5  * For more details take a look at the Java Quickstart chapter in the Gradle
6  * user guide available at https://docs.gradle.org/4.4.1/userguide/tutorial\_java\_projects.html
7  */
8
9 // Apply the java plugin to add support for Java
10 apply plugin: 'java'
11
12 // Apply the application plugin to add support for building an application
13 apply plugin: 'application'
14
15 // In this section you declare where to find the dependencies of your project
16 repositories {
17     // Use jcenter for resolving your dependencies.
18     // You can declare any Maven/Ivy/file repository here.
19     jcenter()
20 }
21
22 dependencies {
23     // This dependency is found on compile classpath of this component and consumers.
24     compile 'com.google.guava:guava:23.0'
25
26     // Use JUnit test framework
27     testCompile 'junit:junit:4.12'
28 }
29
30 // Define the main class for the application
31 mainClassName = 'App'
32 // we can add user defined tasks
33 task display{
34     doLast{
35         println '60 devops class'
36     }
37 }
38
```

Step 6:

To run user defined task execute the commands

```
vaishnavi@vaishnavi-virtualbox:~/devops21/MyGradleApp$ gedit build.gradle  
vaishnavi@vaishnavi-virtualbox:~/devops21/MyGradleApp$ gradle display
```

```
> Task :display  
6D devops class
```

```
BUILD SUCCESSFUL in 1s  
1 actionable task: 1 executed  
vaishnavi@vaishnavi-virtualbox:~/devops21/MyGradleApp$ █
```